UNDER WEAK ASSOCIATION AND LOCALLY FINITE COMPONENTS THE SHEAF OF MEROMORPHIC FUNCTIONS IS THE PRODUCT OF THE SKYSCRAPERS AT THE GENERIC POINTS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Following the multi-claim convention for this source, source_url is the source-level URL https://stacks.math.columbia.edu/ and the source label lemma-meromorphic-weakass-finite is carried in claim_key and here rather than in a per-claim tag URL.

Editorial note. The preamble is the chapter's shared Stacks Project preamble, retained as sources/stacks-divisors/retained-preamble.tex: a line-for-line excerpt of the shared Stacks preamble staged with this snapshot. No macro was invented or redefined, and this card adds no preamble text of its own.

Editorial note. Statement span 6034-6073 and proof span 6075-6132 come from the halfA pool card operations/pool-cards/stacks-divisors-halfA.json and were re-read in the pinned chapter file before this card was written.

Editorial note. The proof contains the author's parenthesis '(argument omitted)' for the extension of the rational-function comparison from the case of finitely many generic points to the general case by glueing; that phrase is kept verbatim and the identity it depends on is proved in the counted span.

Editorial note. The statement carries a long list of numbered consequences; all of them are part of the counted statement span and are preserved verbatim.

Editorial note. 7 cross-references occur inside the counted spans; they are listed in external_references as navigation pointers only.

Allowed standard prerequisites (editorial).

- The description of the total ring of fractions of a ring all of whose weakly associated primes are minimal, and the identification of nonzerodivisors with elements outside the weakly associated primes. Cited at proof lines 6102-6106 and again at 6126-6129. Gives the product decomposition of the total ring of fractions over affine opens and the statement about nonzerodivisors.
- The compatibility of the sheaf of meromorphic functions with tensor products by quasi-coherent modules, and the identification of rational functions with meromorphic functions on an integral scheme. Used at proof lines 6077-6082. Derives the statement about meromorphic sections of an arbitrary quasi-coherent module from the main identity, and handles part of the comparison with rational functions.
- Pullbacks of meromorphic functions along a morphism whose fibres are integral, cited from the same chapter, together with quasi-compactness and quasi-separatedness of the coproduct morphism and quasi-coherence of its

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pushforward of the structure sheaf. Cited at proof lines 6088-6090 and 6120-6123. Produces the comparison map from the sheaf of meromorphic functions to the pushforward and gives its quasi-coherence.

Source reference. Original lemma 02LX: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3424–3424. Same-chapter reference used in the retained author text at source line(s) 6105. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 05C3: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6906–6906. Same-chapter reference used in the retained author text at source line(s) 2624, 6025, 6106, 6127. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 02OU: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3529–3529. Same-chapter reference used in the retained author text at source line(s) 6096, 6992. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original definition 01RT: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2314–2314. Same-chapter reference used in the retained author text at source line(s) 6069. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 01RV: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2316–2316. Same-chapter reference used in the retained author text at source line(s) 1879, 6083. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 01LC: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2081–2081. Same-chapter reference used in the retained author text at source line(s) 741, 6120, 7924. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original section 00Y3: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1241–1241. Same-chapter reference used in the retained author text at source line(s) 6050, 6325, 6435. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Lemma 0.1. *Let X be a scheme such that*

(a) *every weakly associated point of X is a generic point of an irreducible component of X, and*
(b) *any quasi-compact open has a finite number of irreducible components.*

Let X^0 be the set of generic points of irreducible components of X. Then we have

$$\mathcal{K}_X = \bigoplus\nolimits_{\eta \in X^0} j_{\eta,*}\mathcal{O}_{X,\eta} = \prod\nolimits_{\eta \in X^0} j_{\eta,*}\mathcal{O}_{X,\eta}$$

where $j_\eta : \mathrm{Spec}(\mathcal{O}_{X,\eta}) \to X$ is the canonical map of Schemes, Section 00Y3. Moreover

(1) *$\mathcal{K}_X$ is a quasi-coherent sheaf of $\mathcal{O}_X$-algebras,*
(2) *for every quasi-coherent $\mathcal{O}_X$-module $\mathcal{F}$ the sheaf*

$$\mathcal{K}_X(\mathcal{F}) = \bigoplus\nolimits_{\eta \in X^0} j_{\eta,*}\mathcal{F}_\eta = \prod\nolimits_{\eta \in X^0} j_{\eta,*}\mathcal{F}_\eta$$

of meromorphic sections of $\mathcal{F}$ is quasi-coherent,
(3) *$\mathcal{S}_x \subset \mathcal{O}_{X,x}$ is the set of nonzerodivisors for any $x \in X$,*
(4) *$\mathcal{K}_{X,x}$ is the total quotient ring of $\mathcal{O}_{X,x}$ for any $x \in X$,*
(5) *$\mathcal{K}_X(U)$ equals the total quotient ring of $\mathcal{O}_X(U)$ for any affine open $U \subset X$,*
(6) *the ring of rational functions of X (Morphisms, Definition 01RT) is the ring of meromorphic functions on X, in a formula: $R(X) = \Gamma(X, \mathcal{K}_X)$.*

Proof. Observe that a locally finite direct sum of sheaves of modules is equal to the product since you can check this on stalks for example. Then since $\mathcal{K}_X(\mathcal{F}) = \mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{K}_X$ we see that (2) follows from the other statements. Also, observe that part (6) follows from the initial statement of the lemma and Morphisms, Lemma 01RV when X^0 is finite; the general case of (6) follows from this by glueing (argument omitted).

Let $j : Y = \coprod_{\eta \in X^0} \mathrm{Spec}(\mathcal{O}_{X,\eta}) \to X$ be the coproduct of the morphisms j_η. We have to show that $\mathcal{K}_X = j_*\mathcal{O}_Y$. First note that $\mathcal{K}_Y = \mathcal{O}_Y$ as Y is a disjoint union of spectra of local rings of dimension 0: in a local ring of dimension zero any nonzerodivisor is a unit. Next, note that pullbacks of meromorphic functions are defined for j by Lemma 02OU. This gives a map

$$\mathcal{K}_X \longrightarrow j_*\mathcal{O}_Y.$$

Let $\mathrm{Spec}(A) = U \subset X$ be an affine open. Then A is a ring with finitely many minimal primes $\mathfrak{q}_1, \ldots, \mathfrak{q}_t$ and every weakly associated prime of A is one of the $\mathfrak{q}_i$. We obtain $Q(A) = \prod A_{\mathfrak{q}_i}$ by Algebra, Lemmas 02LX and 05C3. In other words, already the value of the presheaf $U \mapsto \mathcal{S}(U)^{-1}\mathcal{O}_X(U)$ agrees with $j_*\mathcal{O}_Y(U)$ on our affine open U. Hence the displayed map is an isomorphism which proves the first displayed equality in the statement of the lemma.

Finally, we prove (1), (3), (4), and (5). Part (5) we saw during the course of the proof that $\mathcal{K}_X = j_*\mathcal{O}_Y$. The morphism j is quasi-compact by our assumption that the set of irreducible components of X is locally finite. Hence j is quasi-compact and quasi-separated (as Y is separated). By Schemes, Lemma 01LC $j_*\mathcal{O}_Y$ is quasi-coherent. This proves (1). Let $x \in X$. We may choose an affine open neighbourhood $U = \mathrm{Spec}(A)$ of x all of whose irreducible components pass through x. Then $A \subset A_{\mathfrak{p}}$ because every weakly associated prime of A is contained in $\mathfrak{p}$ hence elements of $A \setminus \mathfrak{p}$ are nonzerodivisors by Algebra, Lemma 05C3. It follows easily that any nonzerodivisor of $A_{\mathfrak{p}}$ is the image of a nonzerodivisor on a (possibly smaller) affine open neighbourhood of x. This proves (3). Part (4) follows from part (3) by computing stalks. □